Journey Inside the Atom

Chapter 8 | Grade 9 Science (Exploration) | Important Questions with Answers

Section A: Multiple Choice Questions (10 questions)

Q1. Which ancient Indian thinker called the smallest indivisible particles of matter *parmanus*?

(a) Aryabhata
(b) Acharya Kanada
(c) Charaka
(d) Varahamihira

Answer: (b) Acharya Kanada (Vaisesika Sutras).

Q2. John Dalton proposed his atomic theory in the year:

(a) 1803
(b) 1808
(c) 1897
(d) 1911

Answer: (b) 1808. (In 1803 he introduced pictorial symbols for elements.)

Q3. Cathode rays, studied by J. J. Thomson, are streams of:

(a) Protons
(b) Neutrons
(c) Negatively charged particles (electrons)
(d) Alpha particles

Answer: (c) Negatively charged particles called electrons.

Q4. In the gold foil experiment, most alpha particles passed through the foil undeflected. This shows that:

(a) The nucleus is very large
(b) Most of the atom is empty space
(c) Electrons are heavier than protons
(d) The foil was too thick

Answer: (b) Most of the atom is empty space.

Q5. The neutron was discovered in 1932 by:

(a) Rutherford
(b) Bohr
(c) James Chadwick
(d) Thomson

Answer: (c) James Chadwick.

Q6. The maximum number of electrons that the M-shell (n = 3) can hold is:

(a) 8
(b) 2
(c) 18
(d) 32

Answer: (c) 18, since $2n^2 = 2 \times 3^2 = 18$.

Q7. An atom has atomic number 17 and mass number 35. The number of neutrons in it is:

(a) 17
(b) 18
(c) 35
(d) 52

Answer: (b) 18. Neutrons = mass number - atomic number = 35 - 17 = 18.

Q8. The electronic configuration of sodium is 2, 8, 1. Its valency is:

(a) 1
(b) 2
(c) 7
(d) 8

Answer: (a) 1. It loses one electron to get a complete octet.

Q9. Calcium (Z = 20) and argon (Z = 18), both with mass number 40, are examples of:

(a) Isotopes
(b) Isobars
(c) Isomers
(d) Allotropes

Answer: (b) Isobars: different atomic numbers but the same mass number.

Q10. The weighted average atomic mass of chlorine (^{35}Cl : ^{37}Cl = 3 : 1) is:

(a) 36 u
(b) 35 u
(c) 35.5 u
(d) 37 u

Answer: (c) 35.5 u. (35 x 75/100) + (37 x 25/100) = 35.5 u.

Section B: True or False (5 questions)

Q11. The concept of the 'atom' given by Kanada and Democritus came from laboratory experiments.

Answer: False. It originated as an imaginary (philosophical) idea, not from experimental observations.

Q12. Neutrons are present in the nucleus of all atoms except hydrogen (protium).

Answer: True. Protium has one proton and no neutron.

Q13. Isotopes of an element have different chemical properties.

Answer: False. Isotopes have the same number of electrons and the same electronic configuration, so their chemical properties are similar. Only physical properties (like melting and boiling points) differ.

Q14. Rutherford's model could not explain why atoms are stable.

Answer: True. An orbiting electron should lose energy and spiral into the nucleus.

Q15. The outermost shell of an atom can hold a maximum of 8 electrons.

Answer: True. (For the first shell, the maximum is 2.)

Section C: Fill in the Blanks (5 questions)

Q16. The number of protons in the nucleus of an atom is called its _____ number.

Answer: atomic (denoted by Z).

Q17. Protons and neutrons present in the nucleus are together called _____.

Answer: nucleons.

Q18. Thomson's model of the atom is also known as the _____ model.

Answer: plum pudding (or watermelon) model.

Q19. The diameter of an atom is about 10^{-10} m, while the diameter of the nucleus is about _____ m.

Answer: 10^{-15} m (about one lakh times smaller than the atom).

Q20. The shells K, L, M, N were named after the X-ray experiments of the physicist _____.

Answer: Charles Barkla.

Section D: Assertion and Reason (4 questions)

Q21. Assertion (A): Most alpha particles passed straight through the gold foil.
Reason (R): Most of the space inside an atom is empty.

(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, but R is not the correct explanation of A.
(iii) A is true, but R is false.
(iv) A is false, but R is true.

Answer: (i) Both are true and R explains A. Alpha particles go straight because the atom is mostly empty space.

Q22. Assertion (A): Isotopes of an element show similar chemical properties.
Reason (R): Isotopes have the same number of electrons and the same valence electrons.

(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, but R is not the correct explanation of A.
(iii) A is true, but R is false.
(iv) A is false, but R is true.

Answer: (i) Both are true and R explains A. Chemical properties depend mainly on valence electrons.

Q23. Assertion (A): According to Rutherford's model, atoms should collapse.
Reason (R): An electron moving in a circular path is accelerating and should keep losing energy.

(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, but R is not the correct explanation of A.
(iii) A is true, but R is false.
(iv) A is false, but R is true.

Answer: (i) Both are true and R explains A. The electron would spiral into the nucleus.

Q24. Assertion (A): Neon is chemically unreactive.
Reason (R): The atomic number of neon is 10.

(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, but R is not the correct explanation of A.
(iii) A is true, but R is false.
(iv) A is false, but R is true.

Answer: (ii) Both are true, but R is not the correct explanation. Neon is unreactive because its outermost shell has a complete octet (2, 8), not because of its atomic number.

Section E: Short Answer Questions (6 questions)

Q25. What is valency? Give the valency of sodium and chlorine.

Answer: Valency is the number of electrons gained, lost or shared by an atom to complete its octet (its combining capacity). Sodium (2, 8, 1) loses 1 electron, so its valency is 1. Chlorine (2, 8, 7) gains 1 electron, so its valency is 1.

Q26. Write the electronic configuration of magnesium (Z = 12), sulfur (Z = 16) and argon (Z = 18).

Answer: Mg: 2, 8, 2 | S: 2, 8, 6 | Ar: 2, 8, 8.

Q27. Why did Thomson conclude that electrons are present in all atoms?

Answer: He found that the nature of cathode rays was independent of the cathode material and of the gas in the tube. So the particles (electrons) must be a basic component of the atoms of every element.

Q28. Differentiate between isotopes and isobars with one example each.

Answer: Isotopes: atoms of the same element with the same atomic number but different mass numbers, e.g. $^{12}_{6}C$ and $^{14}_{6}C$. Isobars: atoms of different elements with the same mass number but different atomic numbers, e.g. $^{40}_{18}Ar$ and $^{40}_{20}Ca$.

Q29. Bromine exists as ^{79}Br (49.7%) and ^{81}Br (50.3%). Calculate its average atomic mass.

Answer: Average atomic mass = (79 x 49.7/100) + (81 x 50.3/100) = 39.26 + 40.74 = 80.0 u (approximately).

Q30. Why do heavier atoms need many more neutrons than protons in the nucleus?

Answer: All protons carry positive charge and repel each other. Neutrons are neutral, so they reduce this repulsion by increasing the distance between protons, and they strengthen the nuclear force that holds the nucleus together. Example: uranium has 92 protons and 146 neutrons.

Section F: Long Answer Questions (5 questions)

Q31. Describe Rutherford's gold foil experiment. State the observations and the conclusions he drew from it.

Answer:

Experiment (1911, Geiger and Marsden under Rutherford): A narrow beam of alpha particles (positively charged particles from radioactive elements) was aimed at an extremely thin gold foil. Thomson's model predicted that the particles would pass straight through or be deflected only slightly.

Observations: (1) Most alpha particles passed through undeflected. (2) Some were deflected through large angles (scattering). (3) A few bounced back.

Conclusions (planetary / nuclear model): (1) Most of the atom is empty space. (2) The positive charge and most of the mass are concentrated in a tiny, dense centre called the nucleus. (3) Electrons revolve around the nucleus like planets around the Sun. The nucleus is about 10^5 times smaller than the atom.

Limitation: The model could not explain the stability of the atom.

Q32. State the postulates of Bohr's model of the atom. How did it overcome the limitation of Rutherford's model?

Answer:

Postulates (1913): (1) Electrons move only in fixed circular paths called orbits, shells or stationary states. (2) Each shell has a definite energy, so shells are called energy levels; they are named K, L, M, N... or n = 1, 2, 3, 4... (3) While in a fixed shell an electron does not lose energy. (4) The K-shell is nearest to the nucleus and has the least energy; energy increases outward. (5) An electron changes shell by absorbing or releasing energy equal to the difference between the two levels. (6) Each shell can hold only a certain number of electrons.

Overcoming the limitation: Rutherford's electron should lose energy while moving in a circle and fall into the nucleus. Bohr postulated stationary states in which the electron's energy stays constant even though it is in motion, so the atom remains stable.

Q33. Explain Thomson's model of the atom. Why was it rejected?

Answer:

Discovery of the electron (1897): Thomson passed a high voltage through gas at low pressure in a cathode ray tube. Rays moved from cathode to anode. In electric and magnetic fields he showed they were streams of negatively charged particles much lighter than atoms, later called electrons.

Model: Since atoms are neutral, he proposed a sphere of positive charge with electrons embedded throughout it, like plums in a pudding (or seeds in a watermelon). The positive and negative charges balance each other.

Rejection: In the gold foil experiment, some alpha particles were deflected at large angles and a few bounced back. This cannot happen if positive charge is spread evenly, so the model failed to explain the results.

Q34. State the Bohr-Bury rules for distributing electrons in shells. Write the electronic configuration and valency of oxygen and carbon.

Answer:

Rules: (1) The maximum number of electrons in a shell is $2n^2$ (K = 2, L = 8, M = 18). (2) The outermost shell can hold a maximum of 8 electrons (2 for the first shell). (3) Electrons fill shells stepwise, starting from the one closest to the nucleus: K, then L, then M, and so on.

Oxygen (Z = 8): 2, 6. It has 6 valence electrons and gains 2 electrons to complete its octet, so its valency is 2.

Carbon (Z = 6): 2, 4. It has 4 valence electrons and shares 4 electrons with other atoms, so its valency is 4.

Q35. An element X has mass number 35 and contains 18 neutrons. Find (i) protons and electrons, (ii) atomic number, (iii) the element, (iv) electronic configuration, (v) valence electrons, (vi) the mass number if two neutrons are added, (vii) relation with the new atom.

Answer:

(i) Protons = 35 - 18 = 17, so electrons = 17 (the atom is neutral).

(ii) Atomic number = 17.

(iii) The element is chlorine (Cl).

(iv) Electronic configuration: 2, 8, 7.

(v) Valence electrons = 7.

(vi) New mass number = 35 + 2 = 37.

(vii) Both have 17 protons but different mass numbers (35 and 37), so they are isotopes (^{35}Cl and ^{37}Cl).